Baule Speech Dataset For Automatic Speech Recognition Task

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The dataset was created to enable research on automatic speech recognition in Boulé (Baule) language. The dataset was intentionally created with this task in mind, in order to participate in the Google NLP Hack Series: Intro to ASR Africa Challenge hosted on the Zindi Africa platform.
It contains about 565 recordings of participants reading a transcription in Baule as spoken in Côte d'Ivoire, one sentence at a time. Each example contains the audio files and the associated text. The audio is recorded in a less noisy environment by the speakers using their android phone. The dataset is multi-speaker, containing recordings from 4 volunteers (2 males and 2 females), where each volunteer contributed up to 141 recordings. The recordings took place in Abidjan, Côte d'Ivoire in April 2022.

PUBLISHER(S)
data354

INDUSTRY TYPE
Corporate - Tech

KEY APPLICATIONS
Machine Learning, Speech Technology

INTENDED USE CASE(S)
Speech recognition models

PRIMARY DATA TYPE
Speech data

DATASET CHARACTERISTICS

Number of recorded lines	565 lines
Avg. number of lines per participant	141.25 lines
Avg. number of words in script	9.1 words
Number of participants	4 participants
Total length of recordings	50m 26s
Avg. length of recordings	5.3 s
Avg. recording file size	109.4 kB
Human verified instances	all
Recording format	'aac', 'm4a' at 16 kHz, 44 kHz or 48 kHz

NATURE OF CONTENT
The dataset contains recordings of Baule as spoken in Côte d'Ivoire in 2022. The participants read a transcription, approximately one sentence per file. The data is delivered in audio files and the associated transcription of the audio. All the script lines are listed with the corresponding audio files in a file named transcription.tsv, which has two columns. The columns are in order 'id' the file id and 'bci' the Baule text read in the corresponding audio file.

EXAMPLE COMPONENTS
The file line_index.tsv gives a transcription of each audio file in the following format:

Audio: FileID	OD_F01_123
Script: Text	n'klo how
Gender: Text	F or M
Age: Integer	20 to 35 years

DESCRIPTIONS OF EXAMPLE COMPONENTS
OD_F01_123 is the FileID of the file containing the Text in the line. The FileID is composed of three parts, delimited by an underscore "_". The first part is unique for the dataset, the second part is a unique identification of the user, and the third is a unique number for the file.

LICENSE TYPE(S)
Open Access

FIRST RELEASED	April 2022
CURRENT VERSION	Version 1
MAINTENANCE STATUS	actively maintained
CHANGE LOG	N/A
ACCESS LINK	monsia.gougban@data354.co

ATTRIBUTION
Audio datasets for automatic speech recognition in Baule available in open source.

DATA COLLECTION METHOD(S)

Transcription: Web Scraping

DATA SOURCE(S)
Text in Baule from Glosbe online dictionary — Generated by dataset publishers.

DATA COLLECTION PROCEDURE
The dataset set was created in two steps:
• create by scraping a corpus of 14,000 pairs of Baule and French texts on Glosbe online dictionary available on https://app.glosbe.com
• create the audio data from the created text corpus. Currently, 565 audios have been recorded.

DATA SOURCE(S) DESCRIPTION
Glosbe online dictionary available here https://app.glosbe.com/

DATA SELECTION
Lines were randomized and assigned to each user for the recordings.

Audio: Volunteer Collection Efforts

DATA SOURCE(S)
Volunteers — Recording audio of Baule script

DATA SOURCE(S) DESCRIPTION
Volunteers, native Baule speakers located in Ivory Coast. No further demographic information can be reported on the volunteers. Only gender and age information was collected.

DATA SOURCE DISTRIBUTION: GENDER

Female	87.4%
Male	12.6%

SAMPLING METHOD(S)
Transcriptions: Unsampled
Audio: Unsampled

SAMPLING TASK(S)	N/A
SAMPLING DESCRIPTION(S)	N/A

SAMPLING POLICY SUMMARY: N/A

VALIDATION METHOD(S)
Transcriptions: Not validated

VALIDATION TASK(S)	N/A
VALIDATION DESCRIPTION(S)	N/A

VALIDATION POLICY SUMMARY: N/A

	VALIDATOR CHARACTERISTICS	N/A
Audio: Human Verified	VALIDATION TASK(S) VALIDATION DESCRIPTION(S) Validate that the audio files, and double check that the script represent the recorded audio VALIDATOR CHARACTERISTICS The same workers were used for the validation as for the script generation.	Validate the text matches the audio

Data Card Version	1.0